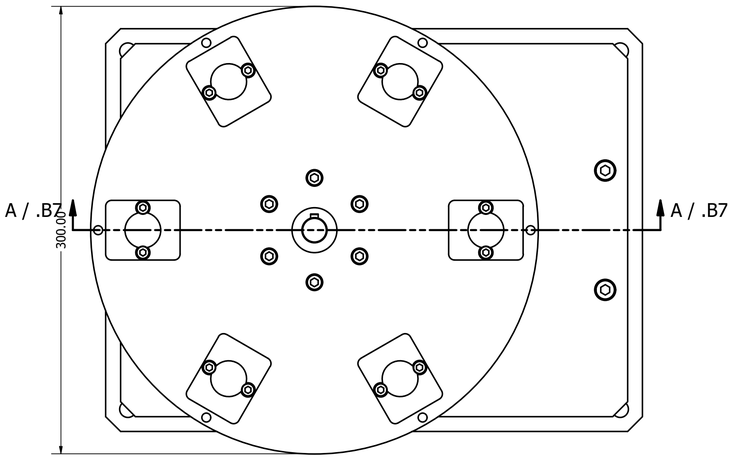
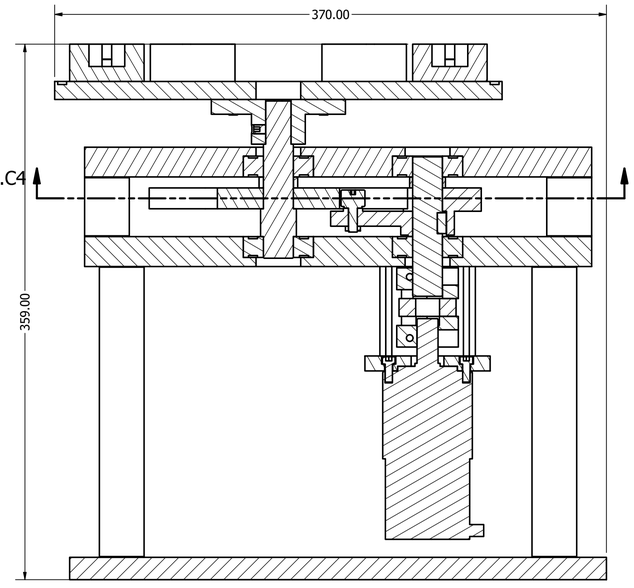
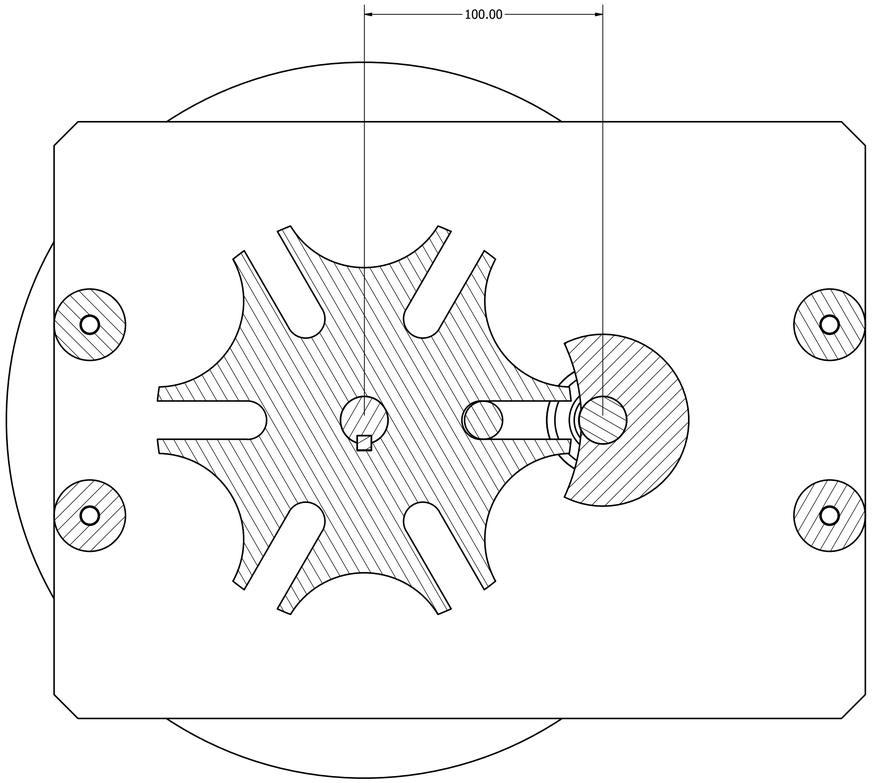
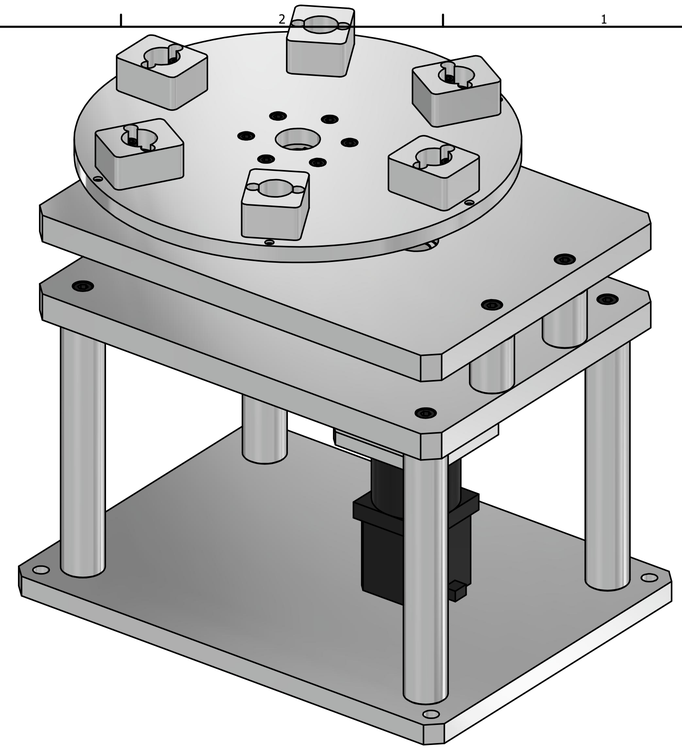




SECTION A-A
SCALE 1 / 2 / .D7

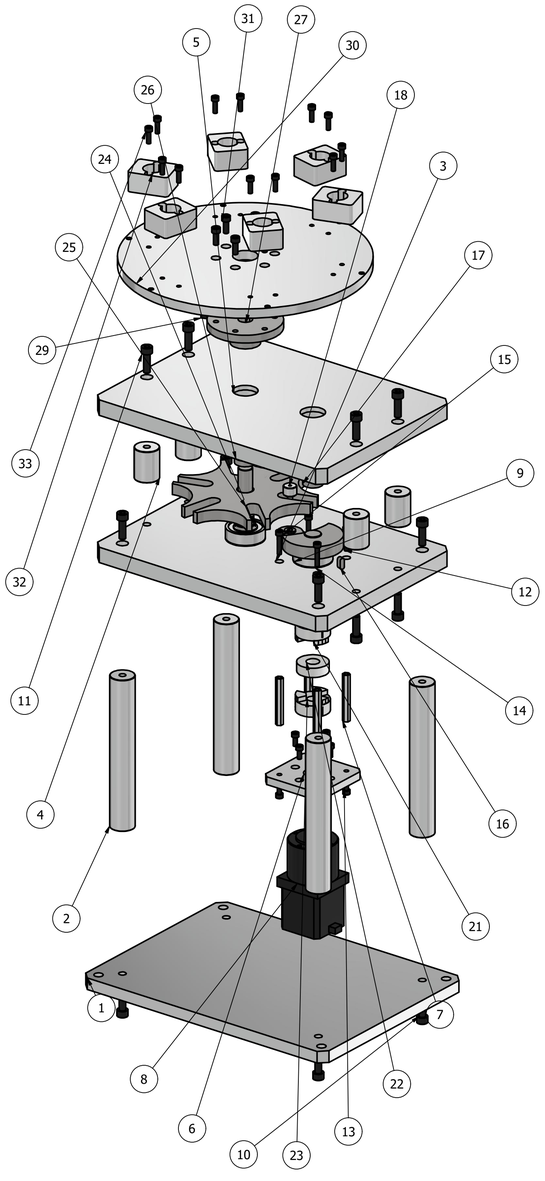


SECTION B-B
SCALE 0.80 : 1 / .B7



GENEVAMECHANISM - 6-STATION ROTARY INDEXING TABLE (DESIGNED AND CALCULATED, NOT BUILT)
6-SLOT EXTERNAL GENEVA, TANGENT ENTRY: C = 100, CRANK a = 50 (= C SIN 30), WHEEL R87, INA KR16-PP ROLLER IN 16.06 SLOTS, LOCKING DISC R36
INDEX 60 DEG PER CRANK TURN: ENGAGED 120 DEG, DWELL 240 DEG (2/3); AT 30 RPM: INDEX 0.667 S, DWELL 1.333 S
DRIVE: ORIENTAL PKP264D14A2-PLE60-5B STEPPER GEARMOTOR 5:1, RULAND MJC41 COUPLING; 4 X SKF 6204-2RSH; 300 X 12 6061 TABLE, 6 NESTS ON R115
CALC.: PEAK CRANK TORQUE 0.87 NM RIGID, 0.89 NM DYNAMIC; MOTOR SF 2.1 ON PULL-OUT (EST.)
LOCK: 0.06 DEG PLAY IN FULL DWELL (0.12 MM AT R115); START / STOP THE CRANK ONLY IN THE DWELL
ASSEMBLY PARAMETER: Crank (0 = MID-INDEX). WHEEL = -(60 TURN + ATAN(0.5 SIN PSI / (1 - 0.5 COS PSI))), PSI CLAMPED TO +-60 DEG.

DRAWN Jay Patel	10/29/2026	
CHECKED		
QA		TITLE: GenevaMechanism - general assembly
MFG		
APPROVED		
		SIZE D · DWG NO GM-DWG-001 · REV
		SCALE 1 / 2 · SHEET 1 OF 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	GM-201	Base plate 6061-T6 360 x 270 x 15: 4 x 9 thru, C'bore 14 x 8.5 from below (M8 into the tall pillars), 4 x 11 bench holes
2	4	GM-204	Pillar 30 x 195 6061-T6, ends faced parallel 0.02, M8 x 20 deep both ends
3	1	GM-202	Lower bearing plate 6061-T6 340 x 250 x 20: 2 x 6204 seats 47 K6 from the top on 100.00 +-0.01 centres, 8 x M8 C'bores (pillars), 4 x M5 C'bores (motor standoffs)
4	4	GM-205	Pillar 30 x 40 6061-T6, ends faced parallel 0.01 (sets the bearing span), M8 x 15 both ends
5	1	GM-203	Upper bearing plate 6061-T6 340 x 250 x 20: 2 x 6204 seats 47 K6 from below, line-bored with the lower plate (dowelled to the short pillars for the boring set-up), 4 x M8 C'bores
6	1	GM-206	Motor plate 6061-T6 84 x 84 x 10: PLE60 pilot 40 H7 x 3.2, 4 x M5 on 52 PCD (gearbox), 4 x M5 C'bore from below on 56 sq (standoffs)
7	4	GM-501	Hex standoff M5 female-female, 8 AF x 60, aluminium (McMaster 94868A class)
8	1	GM-404	Stepper gearmotor Oriental Motor PKP264D14A2-PLE60-5B: NEMA 23 (56.4 mm) 2-phase, 1.4 A, 2.1 ohm, 6 mH, Neugart PLE60 5:1 planetary (10 arcmin), 14 mm shaft; 464 oz-in (3.28 Nm) holding at the output; total length 4.44 in. Envelope model: motor / adapter split and connector assumed
9	4	GM-401	Deep groove ball bearing SKF 6204-2RSH 20 x 47 x 14, sealed: C 13.5 kN, C0 6.55 kN (4: two per shaft, cross-located)
10	4	GM-511	SHCS ISO 4762 M8 x 20, 12.9 (base plate to tall pillars, 4)
11	12	GM-512	SHCS ISO 4762 M8 x 25, 12.9 (bearing plates to pillars, 12)
12	4	GM-516	SHCS ISO 4762 M5 x 25, 12.9 (lower plate to standoffs, 4)
13	8	GM-514	SHCS ISO 4762 M5 x 12, 12.9 (motor plate to gearbox and standoffs, 8)
14	1	GM-103	Crank shaft 20 h6 x 93.5, 1045 CD ground: 2 x 6204 seats, key 6 x 14 for the driver, plain 19 mm end in the MJC41 clamp hub
15	1	GM-102	Geneva driver, one piece 4140 QT: crank arm a = 50 (KR16 stud bore 6 H7), locking disc R36 g6 with R76 relief, hub 36 with 26 bearing land, bore 20 H7 + key 6, M5 set screw
16	2	GM-301	Parallel key DIN 6885 A 6 x 6 x 14 (wheel and driver)
17	1	GM-106	Crank spacer 24 / 20.2 x 7.2: driver to the upper 6204 (24 OD clears the wheel tips by 2 mm)
18	1	GM-402	Stud type track roller (cam follower) INA KR16-PP: D 16, C 11, stud 6 x 16, M6x1, Cr 3200 N, C0r 3450 N (Schaeffler medias); drive roller, nut 3 Nm
19	1	GM-403	Hex nut M6x1 thin (supplied with the KR16-PP), 3 Nm
20	1	GM-517	Set screw ISO 4029 M5 x 6 cup point (driver hub onto the crank shaft, Loctite 243)
21	1	GM-405	Ruland MJC41-20-A jaw coupling hub, 41.3 OD, 20 mm clamp bore (crank shaft)
22	1	GM-407	Ruland MJC41 jaw spider, 98 Shore A urethane (stiffest option); envelope of the spider + jaw overlap
23	1	GM-406	Ruland MJC41-14-A jaw coupling hub, 41.3 OD, 14 mm clamp bore (gearbox shaft)
24	1	GM-104	Wheel / table shaft, 1045 CD: 20 h6 bearing seats, 24 collar (seats the wheel, loads the lower 6204), 17 k6 table-hub seat, keys 6 x 14 and 5 x 16; 105.5 long
25	1	GM-101	Geneva wheel, 6 slots: R87 x 14, slots 16.06 (16 +0.04/+0.08) to r 49, 6 locking arcs R36.1 on 100 centres, bore 20 H7 + key 6; 4140 QT, flanks induction hardened
26	1	GM-105	Wheel spacer 26 / 20.2 x 7.2: wheel to the upper 6204 inner ring (0.1 mm float)
27	1	GM-208	Table hub 1045: bore 17 H7 + key 5, boss 36, flange 90 x 10 with 6 x M6 on 70 PCD, M6 set screw
28	1	GM-302	Parallel key DIN 6885 A 5 x 5 x 16 (table hub)
29	1	GM-518	Set screw ISO 4029 M6 x 8 cup point (table hub onto the shaft, Loctite 243)
30	1	GM-207	Index table 6061-T6 300 x 12, faced flat 0.05: bore 30, 6 x M6 C'bore on 70 PCD, 6 stations on R115 (2 x M5 each), station marks
31	6	GM-513	SHCS ISO 4762 M6 x 16, 12.9 (table to hub, 6)
32	6	GM-209	Station nest 6061-T6 50 x 40 x 25 with a 24 H8 x 15 part pocket (example workpiece seat; replace per job), 2 x M5
33	12	GM-515	SHCS ISO 4762 M5 x 16, 12.9 (station nests, 12)

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CHECKED		TITLE: GenevaMechanism - general assembly
QA		
MFG		
APPROVED		SIZE D | DWG NO GM-DWG-001 | REV
		SCALE 0.29 : 1 | SHEET 2 OF 2